PHASE 2 REPORT – Exploratory Data Analysis (EDA)

🟦 1. Title

Phase 2: Exploratory Data Analysis (EDA) and Data Cleaning

🟦 2. Objective

The objective of this phase is to perform exploratory data analysis on the dataset and handle missing or inconsistent values to make the data suitable for further processing and modeling.

🟦 3. Introduction

Exploratory Data Analysis (EDA) is a crucial step in any data science project. It helps in understanding the structure, patterns, and quality of the dataset.
In this phase, we analyze the dataset, handle missing values, and convert categorical data into numerical form for better analysis.

🟦 4. Dataset Description

The dataset contains logistics-related information such as:

- Average Speed (km/h)
- Delivery Time (minutes)
- Traffic Level
- Weather Impact

This data is used to analyze delivery performance and external factors affecting logistics.

🟦 5. Tools & Technologies Used

- Python
- Jupyter Notebook / Google Colab
- Libraries:
    - pandas
    - numpy

🟦 6. Data Cleaning

(a) Handling Missing Values

Missing values were handled using the following methods:

- Numerical columns (Avg_Speed_kmph, Delivery_Time_min) were filled using **mean values**
- Categorical columns (Traffic_Level, Weather_Impact) were filled using **mode values**

This ensures no data loss while maintaining dataset consistency.

(b) Encoding Categorical Data

The categorical feature **Traffic_Level** was converted into numerical values:

- Low → 1
- Medium → 2
- High → 3

This transformation allows machine learning algorithms to process the data effectively.

🟦 7. Exploratory Data Analysis

(a) Dataset Information

- Checked data types and structure using info()
- Verified non-null values

(b) Statistical Summary

- Used describe() to analyze:
    - Mean
    - Standard deviation
    - Minimum and maximum values

(c) Null Value Analysis

- Verified missing values using isnull().sum()
- Ensured dataset is clean after preprocessing

🟦 8. Implementation

The dataset was processed using Python, where missing values were handled and categorical variables were encoded.
Basic EDA functions were applied to understand the dataset structure and statistics.

🟦 9. Output

- Clean dataset with no missing values
- Numerical representation of categorical data
- Summary statistics generated
- Dataset ready for further modeling

🟦 10. Conclusion

In this phase, we successfully cleaned and analyzed the dataset using EDA techniques. The dataset is now structured, consistent, and ready for advanced analysis such as clustering or prediction in the next phase.

🟦 11. Future Scope

In the next phase, machine learning algorithms like clustering (K-Means) will be applied to identify patterns and optimize logistics routes.